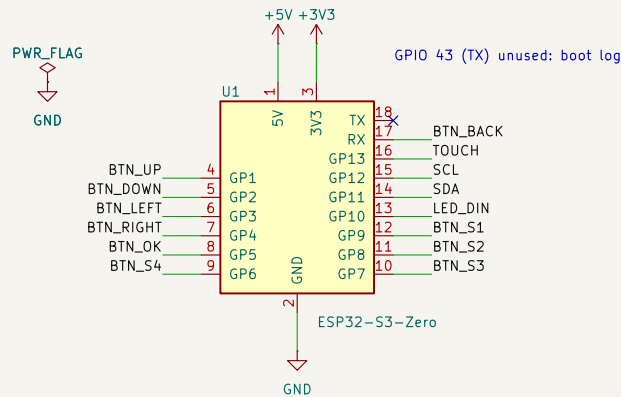


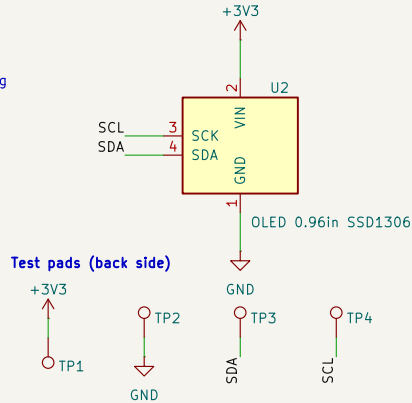
ESP32-S3-Zero module and power

+5V (VBUS) for the LEDs only; +3V3 from the module regulator for the OLED only



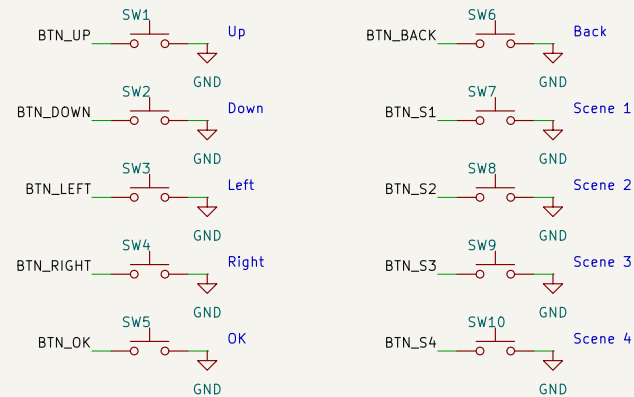
I²C bus and display

OLED 0x3C, pull-ups on the module



Buttons and touch area

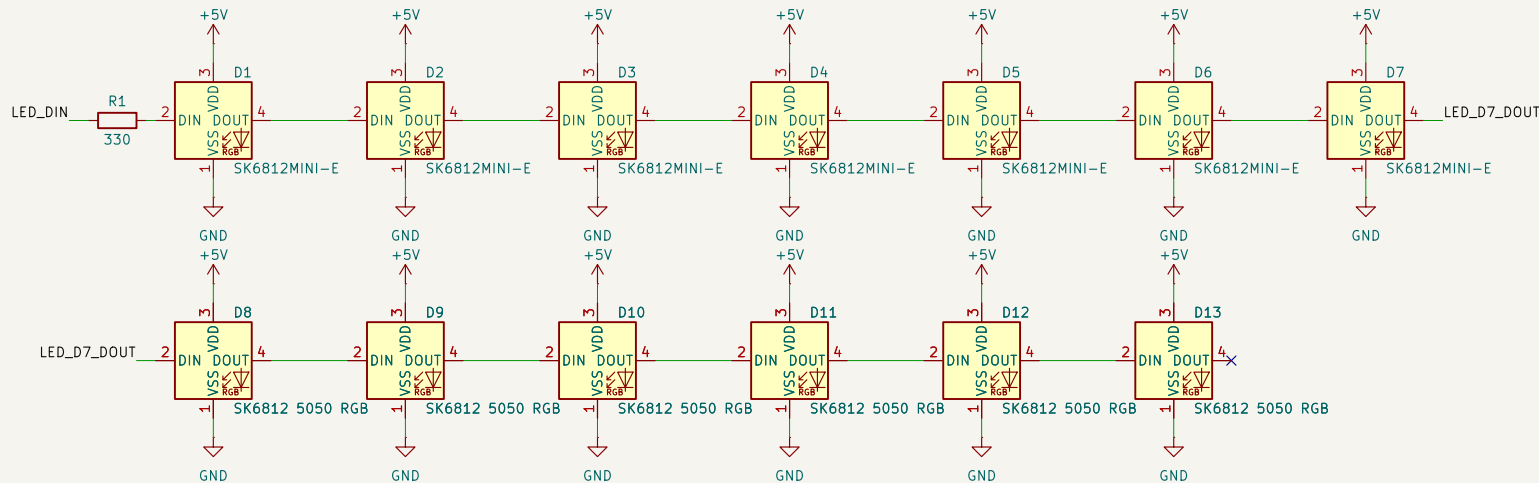
Buttons to GND, ESP32 internal pull-ups



E1: Ø 6 mm copper disc under solder mask, native touch on GPIO 13

LED chain (+5V): D1-D3 status, D4-D7 scenes (SK6812MINI-E), D8-D13 backlight (SK6812 5050)

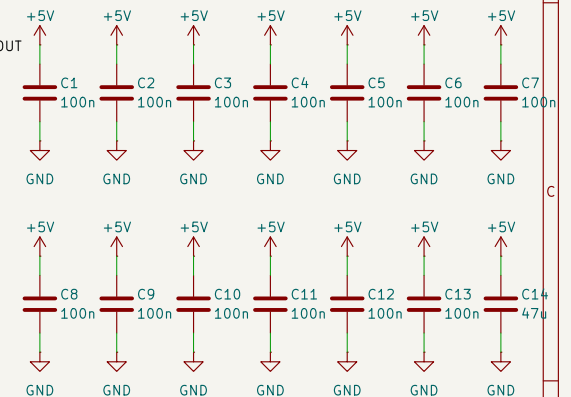
All mounted on the back; chain order = numbering, firmware index = number - 1



Decoupling

C1-C13 100 nF: Cn next to Dn, < 2 mm from VDD

C14 47 µF 10 V bulk, close to D1 and R1



Waveshare ESP32-S3-Zero, SSD1306 OLED, 10 buttons, touch, 13 SK6812 LEDs
Desk button panel for Home Assistant (ESPHome)

Sheet: /
File: planch.kicad_sch

Title: Planch

Size: A4 Date: 2026-10-03

KiCad E.D.A. 10.0.6

Rev: 1

Id: 1/1